

Total No. of Questions : 10]

SEAT No. :

P1960

[Total No. of Pages : 3

[5059] - 542
B.E. (Mechanical)
POWER PLANT ENGINEERING
(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4 Q.5or Q.6, Q.7 or Q. 8, Q.9 or Q. 10.*
- 2) Figures to the right indicate full marks.*
- 3) Use of scientific calculator is allowed.*
- 4) Assume data wherever necessary and mention it.*
- 5) Draw neat and suitable figures wherever necessary.*

- Q1)** a) Discuss the present scenario of power generation in India. **[4]**
b) During trial on steam condenser following observations were recorded.
Condenser vacuum = 680 mm of Hg
Barometer reading = 764 mm of Hg
Mean condenser temp. = 36.2°C
Hot well temp. = 30 °C
Condensate formed per hour = 1780 Kg
Circulating cooling water inlet temp. = 20°C
Circulating cooling water outlet temp. = 32°C
Quantity of cooling water = 1250 Kg/min.
Determine i) condenser vacuum corrected to standard barometer ii) vacuum efficiency iii) under cooling of condensate iv) condenser efficiency. **[6]**

OR

- Q2)** a) From following data calculate cost of generation per unit delivered from the power plant. **[6]**
Installed capacity of power plant = 200 Mw
Annual load factor = 0.4
Capital cost of power plant = Rs. 280 Lacks
Annual cost of fuel, Oil, Salaries & taxation = Rs. 60 Lacks
Interest & depreciation = 13%
b) With neat sketch explain ash handling system. **[4]**

P.T.O.

- Q3) a)** Write a note on Hydrograph and Flow duration curve. [4]
- b) Explain with neat sketch CANDU reactor. Also state the merits and demerits. [6]

OR

- Q4) a)** Discuss the site selection, advantages & disadvantages for Hydroelectric Power Plant. [6]
- b) Write a note on Nuclear waste disposal. [4]
- Q5) a)** Draw a typical layout of Diesel Power Station and Explain it. [6]
- b) A Gas turbine unit has pressure ratio 6:1 & maximum cycle temperature of 610°C. The isentropic efficiency of the compressor and turbine are 0.80 and 0.82 respectively. Calculate power output in kilowatts of an electrical generator geared to the turbine when the air enters the compressor at 15°C at the rate of 16 Kg/Sec. Also calculate thermal efficiency and work ratio of the plant. Assume i) $C_p = 1.005 \text{ Kj/Kg}$ & $\gamma = 1.4$ for compression processes ii) $C_p = 1.11 \text{ Kj/Kg}$ & $\gamma = 1.333$ for expansion processes iii) $C_p = 1.11 \text{ Kj/Kg}$ for combustion processes. [10]

OR

- Q6) a)** Discuss the selection of engine size, advantages & disadvantages of Diesel engine power plant? [6]
- b) In a Gas turbine the compressor takes in air at a temp. of 15°C and compresses it to four times the initial pressure with an isentropic efficiency of 82%. The air is then passed through heat exchanger heated by turbine exhaust before reaching combustion chamber. In the heat exchanger 78% of available heat is given to air. The maximum temp. after constant pressure combustion is 600°C and efficiency of turbine is 70%. Neglecting all losses except those mentioned and assuming working fluid throughout the cycle to have characteristics of air. Find the thermal efficiency of cycle. Assume : $R=0.287 \text{ Kj/Kg.K}$ and $\gamma = 1.4$ for air and constant specific heat throughout. [10]

- Q7)** a) Write a note on
- i) Challenges in commercialization of Non-Conventional Power Plants. [5]
 - ii) Low & High Temperature Solar Power Plant. [5]
- b) Explain single basin and double basin tidal power plant with neat diagrams. [8]

OR

- Q8)** a) With neat sketch explain the working of a fuel cell and list out its advantages over other nonconventional systems. [10]
- b) Discuss the parameters to be consider for site selection of wind power plant. [8]
- Q9)** a) State the various protective equipments and explain the working of switch gear in power plant. [8]
- b) Write a note on thermal pollution of water & its control. [4]
- c) Write a note on Noise pollution by power plants. [4]
- Q10)** a) Draw the typical layout of electrical equipment & discuss the role of control room. [8]
- b) What are different methods used to control SO_2 in the flue gases? Explain any two. [8]

